Sayantani Bhattacharya

Department of Mathematics and Statistics, Auburn University, AL, 36849

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Education

Auburn University, Alabama, USA *Aug 2023 – Present*
PhD in Mathematics

Tezpur University, Assam, India *2018 – 2020*
Master's in Mathematics

Chennai Mathematical Institute, Chennai, India *2015 – 2018*
Bachelor's in Mathematics and Computer Science (Dual Degree)

Research Interests

Matroid Theory, Combinatorics

Preprints & Publications

- S. Bhattacharya, J. D. Clifton, Z. Walsh, The unbreakable quasi-graphic matroids. **Submitted**. [arXiv]

Talks

- **The Unbreakable Quasi-graphic Matroids**
  34th Cumberland Conference on Combinatorics, Graph Theory and Computing, Auburn University, Auburn, AL May 17, 2026
- **The Unbreakable Quasi-graphic Matroids**
  Combinatorics Seminar, Auburn University, Auburn, AL Apr 8, 2026

Teaching Experience

Graduate Teaching Assistant, Auburn University, AL

- MATH 2630 (Calculus III), Sections R10, R11: Recitation Leader Fall 2026
- MATH 1610 (Calculus II), Sections R25, R26: Recitation Leader Spring 2026
- MATH 1950 MR1 (Calculus I, Merit Recitation Program): Instructor of Record Fall 2025
- MATH 1610 (Calculus I), Sections R15, R16: Recitation Leader Spring 2025
- MATH 1000 (College Algebra), Sections 005, 006: Instructor of Record Fall 2024
- MATH 1000/1120/1130/1150/2650: Tutoring Center GTA Summer 2024
- MATH 1610 (Calculus I): Grader Spring 2024

◇ MATH 2650 (Linear Differential Equations): Grader Fall 2023

Graduate Research Assistant, Auburn University, AL

◇ Research Assistant (funded by Dr. Zachary Walsh) Summer 2026

◇ Merit Recitation Program (funded by Dr. Melinda Lanius and Dr. Valentin Küchle) Summer 2025

Fellowships & Awards

◇ $1000 Professional Development Award for completing the Auburn University Summer STEM Institute (AUSSI) Training Program 2025

◇ Graduate Teaching Assistantship and Fellowship, Auburn University 2023 – Present

◇ CSIR India National Eligibility Test (NET) with Junior Research Fellowship (JRF) Nov 2019

◇ INSPIRE Scholarship for Higher Education (SHE) Fellowship, Govt. of India 2015 – 2018

◇ Institutional Fellowship for Undergraduates, Chennai Mathematical Institute 2015 – 2018

Workshops & Field Schools

◇ **Professional Development for Emerging Education Researchers (PEER) Institute Field School**, Auburn University Jun 2026

◇ **Science Communication Workshop Series**, Biggio Center, Auburn University Fall 2025

◇ **Topological Methods in Combinatorics Summer School**, Iowa State University May 2025

◇ **Advanced Instructional School on Advanced Linear Algebra (AIS-ALA 2022)**, ATM Schools, National Center for Mathematics, Mumbai, India May 2022

Service

◇ **Association for Women in Mathematics (AWM) at Auburn University**

- ▷ President 2026 – 27
- ▷ Treasurer 2025 – 26

◇ Secretary and 4th year Cohort Representative, **Mathematics and Statistics Graduate Student Council (MSGSC)**, Auburn University 2026 – 27

◇ DMS Volunteer, **34th Cumberland Conference**, Auburn University May 15–17, 2026

◇ Mentor, **First Year Experience (FYE) Mentorship Program**, Graduate Student Council, Auburn University 2025 – 26

- ▷ *Awarded the Outstanding First Year Experience (FYE) Mentor Award 2025 by the Auburn University Graduate School.*

◇ President, **College of Science & Mathematics International Students Association (CISA)**, Auburn University Nov 2023 – Aug 2024

Outreach

- **Mathematical Puzzle Programs (MaPP)**, Auburn University
  - Volunteer, Feb 21st, 2026
  - Volunteer, Feb 15th, 2025
- **Alabama State Science Olympiad**, Auburn University
  - Volunteer, Bungee Drop Div C, Feb 14th, 2026
  - Volunteer, Codebusters Div B, Mar 29th, 2025
- **Destination STEM Expo**, Auburn University
  - Exhibitor (Chess game : Rook's Move; Puzzle game : Gerrymandering), Oct 1st, 2026
  - Exhibitor (Model : Brachistochrone Curve), Oct 15th, 2025
- Volunteer, **Julia Robinson Mathematics Festival (JRMF) at Joint Mathematics Meeting (JMM)**, Washington DC (Jan 4th, 2026)

Technical Skills

Programming: SageMath, Python, C, LaTeX, HTML
Professional: Microsoft Office, Canva, Adobe (InDesign, Lightroom, Photoshop)